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Blume–Capel model in $d = 1$ with long-range interactions: Giant reentrance in the finite-temperature tricritical phase diagrams[☆]

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ABSTRACT

The Blume–Capel model in one dimension with long-range power-law interactions is studied by renormalization-group theory. A series of finite-temperature tricritical phase diagrams is found, as a function of the power-law exponent of the power-law interactions. These calculated phase diagrams exhibit a giant reentrance in the form disorder-order–disorder as temperature is lowered. The first-order transition takes over the entire phase boundary at longest-range interactions, as a near-equivalent-neighbor regime is approached.

1. Introduction: Power-law Blume–Capel

Microscopic model systems have played a crucial role in understanding and predicting the complex mathematics of nature, such has the critical singularities at a phase transition (indeed, the occurrence of a phase transition) and the chaos inside a spin-glass phase. H. W. Capel [1], together with M. Blume [2], have made a seminal contribution to the usefulness of model systems, by including to the local degree of freedom, the magnitude as well as the alignment, leading to a multitude of successful theories for qualitatively new phenomena.

The one-dimensional Blume–Capel model with finite-range interactions shows no finite-temperature ordering and phase transitions. However, we see below that, with power-law decaying long-range interactions, finite-temperature ordering with first- and second-order phase transitions, indeed a multiplicity of tricritical phase diagrams and unexpectedly giant reentrances, in the form of disorder followed by order followed by disorder as temperature is lowered, occur.

We study a Blume–Capel model with long-range power-law interactions, defined by the power-law long-ranged Hamiltonian

$$-\beta\mathcal{H} = \sum_{r_1 \neq r_2} [J\, s_{r_1} s_{r_2} - \Delta(s_{r_1}^2 + s_{r_2}^2)]\, |r_1 - r_2|^{-a} \tag{1}$$

where $\beta = 1/k_B T$ is the inverse temperature, r_1 and r_2 designate the sites on the one-dimensional system, at each site r_i there is a spin $s_{r_i} = \pm 1, 0$, and the sums are over all sites in the system. The interaction is ferromagnetic, $J > 0$. In the second term, Δ is the chemical potential of the non-zero spins.

It is generally known that phase transitions are absent in strictly one-dimensional systems with short-range interactions, due to the Landau argument balancing local energy disruption versus the entropy of the position of the disruption. For long-range

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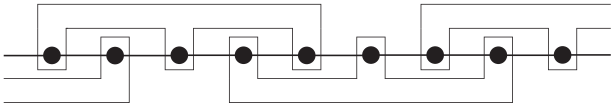

Fig. 1. Renormalization-group cells for $d = 1$.

interactions, the energy disruption may overcome the entropy and finite-temperature ordering happens in one dimension, as seen here.

Other recent similar works are in [3,4].

2. Renormalization-group theory

We study this system by renormalization-group theory, using the 3-spin cells shown in Fig. 1. The three site-spins inside each cell determine a cell-spin $s'_{r'_i} \pm 1, 0$, where primes are used to designate quantities of the renormalized system: The majority value of the site-spins is the value of the cell-spin. When no majority exists, meaning that the site-spins have the values $(+1, -1, 0)$ in any order, the cell-spin has the value 0. This reflects the fact that local disorder represents an effective vacancy and in fact has been used to determine the crossover from second-order to first-order phase transitions in Potts models [5,6] and the multicritical phase diagrams of the Ashkin-Teller models [7,8].

The renormalized interactions are obtained from a partial summation of the partition function Z,

$$\begin{aligned} Z = \sum_{\{s\}} e^{-\beta\mathcal{H}(\{s\})} &= \sum_{\{s'\}} \sum_{\{\sigma\}} e^{-\beta\mathcal{H}(\{s'\},\{\sigma\})} \\ &= \sum_{\{s'\}} e^{-\beta\mathcal{H}'(\{s'\})} = Z', \end{aligned} \tag{2}$$

where the summed variable σ represents, for each cell, the seven (in the case of $s'_{r'_i} = \pm 1$) or thirteen (in the case of $s'_{r'_i} = 0$) site-spin states that give the same cell-spin value. Thus, extracting from the latter equation,

$$e^{-\beta\mathcal{H}'(\{s'\})} = \sum_{\{\sigma\}} e^{-\beta\mathcal{H}(\{s'\},\{\sigma\})}, \tag{3}$$

yields the renormalized interactions. Under the renormalization-group transformations, the long-ranged Hamiltonian takes the most general form [9,10]

$$-\beta\mathcal{H} = \sum_{r_1 \neq r_2} [J_{r_1 r_2} s_{r_1} s_{r_2} + K_{r_1 r_2} s_{r_1}^2 s_{r_2}^2 - \Delta_{r_1 r_2}(s_{r_1}^2 + s_{r_2}^2)]. \tag{4}$$

Eq. (1) gives the initial conditions of the renormalization-group flows that are in terms of Eq. (4).

3. Niemeyer–van Leeuwen 2-cell cluster

Niemeyer and van Leeuwen's two-cell cluster approximation [10–12] is used to solve this system. The two-cell cluster approximation consists in carrying our transformation of Eq. (3) for two cells, including the 6 intracell interactions and the 9 intercell interactions. A recursion relation is obtained for each renormalized interaction, for each (r_1, r_2),

$$\begin{aligned} J' &= \frac{1}{2} \ln \frac{R(+1,+1)}{R(+1,-1)}, \quad \Delta' = \ln \frac{R(+1,0)}{R(0,0)}, \\ K' &= \frac{1}{2} \ln \frac{R(+1,+1)R(+1,-1)R(0,0)^2}{R(0,+1)^4}, \end{aligned} \tag{5}$$

where

$$R(s'_{r'_1}, s'_{r'_2}) = \sum_{\sigma_{r'_1}, \sigma_{r'_2}} e^{-\beta\mathcal{H}_{r'_1 r'_2}}, \tag{6}$$

where the latter unrenormalized two-cell Hamiltonian contains the 6 intracell interactions and the 9 intercell interactions between the 6 spins in cells at r'_1 and r'_2.

Our (approximate) renormalization-group calculation proceeds as follows: We start with a linear system of $3^8 = 6561$ spins, all interacting with the starting long-range power-law interaction given in Eq. (1). After one renormalization-group transformation, we have 2187 spins, all interacting with the renormalized long-range interaction shown in Eq. (4). After one more renormalization-group transformation, we have 729 spins, all interacting with the once more renormalized long-range interaction shown in Eq. (4). We can thus viably carry out 6 renormalization-group transformations. For spot checks, we start with $3^{12} = 531,441$ spins and can carry out 10 transformations. It is thus seen that our calculation becomes less viable for longer-ranger interactions, namely smaller a.

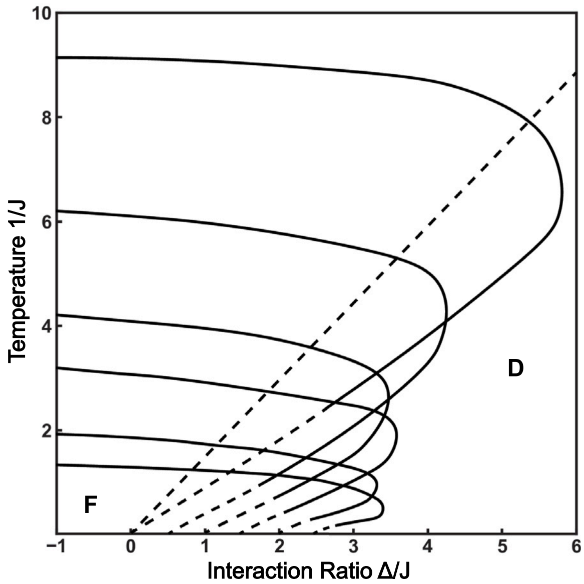

Fig. 2. Calculated phase diagrams of the $d = 1$ Blume–Capel model with long-range power-law interactions. Ferromagnetic (F) and disordered (D) phases are seen. First- and second-order phase boundaries are respectively shown with dashed and full lines. The dashed and full lines meet at a tricritical point. The calculated phase diagrams are, from high temperatures downwards, for $a = 0.5, 0.85, 0.9, 1.0, 1.1, 1.3, 1.5$. After the $a = 0.85$ phase diagram, each consecutive phase diagram is drawn shifted by 0.5 in Δ/J, for better visibility. Thus, in fact the first-order boundaries of all phase diagrams meet at $\Delta/J = 0$ at zero temperature $1/J = 0$. As examplified in this figure, for $a > 0.74$, the phase diagrans show a giant (numerically speaking) reentrance behavior, as the sequence disordered to ferromagnetic to disordered as temperature is lowered. The system for $a = 0.5$ is in the near-equivalent-neighbor regime: In the absence of zeroes, this system is magnetized for all non-infinite temperatures. However, with the introduction of vacancies, the lower branch of the reentrant phase diagram occurs, with a first-order reverse phase transition from ordered phase to disordered phase as temperature is lowered, as seen in this figure. The critical temperatures for no zeroes $\Delta/J \to -\infty$ and the tricritical temperatures, as a function of the interaction-range exponent a, are shown in Fig. 3.

4. Long-range Blume–Capel tricritical phase diagrams in $d = 1$

7 Different calculated phase diagrams of the $d = 1$ Blume–Capel model with long-range power-law interactions are given overlaid in Fig. 2. Ferromagnetic (F) and disordered (D) phases are seen. First- and second-order phase boundaries are respectively shown with dashed and full lines. The dashed and full lines meet at a tricritical point. Starting at a point in each phase, the renormalization-group trajectory flows to a completely stable fixed point, which is the sink of that phase [10]. Thus, the whole basin of attraction of a sink constitutes the corresponding thermodynamic phase. The sinks are best characterized by the exponentiated Hamiltonian between two interacting sites,

$$\mathbf{T}_{ij} \equiv e^{-\beta \mathcal{H}_{r_i r_j}} = e^{J_{ij} s_i s_j + K_{ij} s_i^2 s_j^2 - \Delta_{ij}(s_i^2 + s_j^2)} =$$
$$\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{matrix} \qquad \begin{matrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{matrix} \qquad \begin{matrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{matrix}\,, \tag{7}$$

where each matrix has been divided to its largest element. The first matrix is the sink of the ferromagnetic phase. The last two matrices are the sinks of the dense and dilute disordered phases. These matrices occur, at the sinks, for any two spins at any separation.

Starting at a point on each phase boundary, the renormalization-group trajectory flows to a singly unstable fixed point, which is unstable (in opposite directions) towards the two phases that are separated by this boundary. Thus, in the present case,

$$\mathbf{T}_{ij} = \begin{matrix} 1 & 0.79 & 0 \\ 0.79 & 1 & 0 \\ 0 & 0 & 0 \end{matrix} \qquad \begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix} \qquad \begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{matrix}\,, \tag{8}$$

which are the attractors of the phase boundaries between the ferromagnetic and dense disordered, ferromagnetic and dilute disordered, dense and dilute disordered phases, which respectively give second-order, first-order, no-phase-transition boundaries. Thus, the entire global phase diagram is extracted from the renormalization-group flows.

The calculated phase diagrams in Fig. 2 are, from high temperatures downwards, for $a = 0.5, 0.85, 0.9, 1.0, 1.1, 1.3, 1.5$. In the sequence of phase diagrams, the transition temperatures are lowered as the power-law exponent a of the interactions is increased,

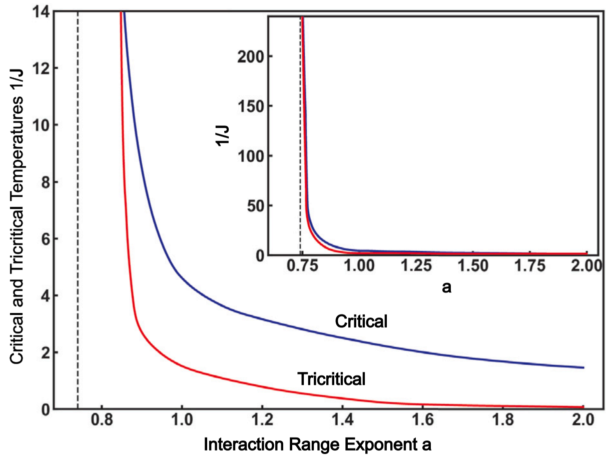

Fig. 3. The critical temperatures in the no-zeros limit, $\Delta/J \to -\infty$, and the tricritical temperatures (lower curve), as a function of the interaction-range exponent a. A finite-temperature ferromagnetic phase transition occurs for $0.74 < a < 2$. The second-order phase transition temperature monotonically decreases between these two limits. For $a = 2$ the phase transition temperature discontinuously drops to zero and for $a > 2$ there is no ordered phase above zero temperature, as predicted by rigorous results [13–18]. At the other end, on approaching $a = 0.74$ from above, the phase transition temperature and the tricritical temperature both diverge to infinity, meaning that, at all non-infinite temperatures, the system is ferromagnetically ordered. The first-order transition takes over the entire phase boundary as this limit is approached. Thus, the equivalent-neighbor-interactions regime is entered before ($a > 0$) the neighbors become equivalent, namely before the interactions become equal for all separations. To the left of the dashed line on this figure is the near-equivalent-neighbor regime which occurs here as an approximate result at $a = 0.74$, whereas the exact onset is at $a = 1$.

as expected, since this is in the direction of the short-range interaction limit of a going to infinity. With the introduction of vacancies (positive Δ), a condensation of vacancies precipitates first-order phase transitions [3,4]. Reentrance suggests that, at intermediate temperatures where the ferromagnetic phase would be unfavorable, vacancies stabilize the ferromagnetic phase by bounding, and therefore delimiting, the opposite spin regions.

In Fig. 2, after the $a = 0.85$ phase diagram, each consecutive phase diagram is drawn shifted by 0.5 in Δ/J, for better visibility. Thus, in fact the first-order boundaries of all phase diagrams meet at $\Delta/J = 0$ at zero temperature $1/J = 0$. As examplified in this figure, for $a > 0.74$, the phase diagrams show a giant (numerically speaking) reentrance behavior [19–26], as the sequence disordered to ferromagnetic to disordered as temperature is lowered. The system for $a = 0.5$ is in the near-equivalent-neighbor regime. As seen in Fig. 2, in the absence of zeros, in this regime this system is magnetized for all non-infinite temperatures. However, with the introduction of zeros via the chemical potential, a first-order phase diagram occurs, with a reverse phase transition, from high-temperature ordered to low-temperature disordered phase, as seen in this figure. This is the remaining lower branch of the reentrant phase diagram.

The critical temperatures in the no-zeros limit, $\Delta/J \to -\infty$, and the tricritical temperatures, as a function of the interaction-range exponent a, are given in Fig. 3. A finite-temperature ferromagnetic phase transition occurs for $0.74 < a < 2$. The second-order phase transition temperature monotonically decreases between these two limits. At $a = 2$ the phase transition temperature discontinuously drops to zero, as predicted by rigorous results [16,17], and for $a > 2$ there is no ordered phase above zero temperature, also as predicted by rigorous results [14,15]. At the other end, on approaching $a = 0.74$ from above, the phase transition temperature and the tricritical temperature both diverge to infinity, meaning that, at all non-infinite temperatures, the system is ferromagnetically ordered. The first-order transition takes over the entire phase boundary as this limit is approached. Thus, the equivalent-neighbor-interactions regime is entered before ($a > 0$) the neighbors become equivalent, namely before the interactions become equal for all separations. To the left of the dashed line on this figure is the near-equivalent-neighbor-regime which occurs here as an approximate result at $a = 0.74$, whereas the exact onset is at $a = 1$ [18]. In the equivalent-neighbor limit, a phase transition occurs in the infinitesimal interaction limit [27].

5. Conclusion

We solved the Blume–Capel model in one dimension with long-range power-law interactions by renormalization-group theory. We found a series of finite-temperature tricritical phase diagrams, as a function of the power-law exponent of the power-law interactions. These calculated phase diagrams exhibit a giant reentrance in the form disorder-order–disorder as temperature is lowered. One branch of the tricritical phase diagram also occurs, as a reverse transition, namely order-to-disorder as temperature is lowered, in the longer-range of interactions, in the near-equivalent-neighbor regime of the system. The first-order transition takes over the entire phase boundary at longest-range interactions, as a near-equivalent-neighbor regime is approached. It is clear thus that a variety of other complex finite-temperature phenomena can be studied on this system, further expanding the legacy of Blume and Capel.

Furthermore, our method can readily be extended and new phenomena can be studied in complex systems, such as chaotic spin glasses, nonrandom but competing interactions, and in fact quantum systems where variable-range hopping can be implemented for the tJ model for high-temperature superconductivity [28,29].

CRediT authorship contribution statement

E. Can Artun: Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **A. Nihat Berker:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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